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BAND SAW

Abstract

A benchtop band saw including a base, a clamp assembly configured to hold a workpiece, a support arm pivotably coupled to the base about a miter axis extending perpendicular to the base, and a band saw moveable relative to the support arm and the base between a raised position and a lowered position for performing a cutting operation. The band saw has a housing defining an opening in which the workpiece is partially receivable, a motor, a first wheel supported by the housing for rotation about a first wheel axis, a second wheel supported by the housing for rotation about a second wheel axis, and a band saw blade. The miter axis is located within a reference plane oriented parallel to the first wheel axis and the second wheel axis. The reference plane extends through the opening when the band saw is in at least the lowered position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/654,455, filed May 31, 2024, and U.S. Provisional Patent Application No. 63/560,176, filed Mar. 1, 2024, the entire contents of both of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to power tools, and more specifically to band saws.

BACKGROUND OF THE INVENTION

[0003] Benchtop band saws are used to cut a variety of workpieces, such as pipes, rods, studs, and blocks. Benchtop band saws typically include a base and a band saw pivotably coupled to the base for cutting the workpieces.

SUMMARY OF THE INVENTION

[0004] In some aspects, the techniques described herein relate to a benchtop band saw including: a base; a clamp assembly coupled to the base and configured to hold a workpiece; a support arm pivotably coupled to the base about a miter axis that extends perpendicular to the base; and a band saw coupled to the support arm and moveable relative to the support arm and the base between a raised position and a lowered position for performing a cutting operation on the workpiece, the band saw including a housing defining an opening in which the workpiece is at least partially receivable during the cutting operation, a motor positioned within the housing, a first wheel supported by the housing for rotation about a first wheel axis, the first wheel being coupled to the motor to receive torque therefrom, a second wheel supported by the housing for rotation about a second wheel axis, and a band saw blade extending around the first wheel and the second wheel and traveling in a continuous loop around the first wheel and the second wheel, the band saw blade traversing the opening in the housing, wherein the miter axis is located within a reference plane oriented parallel to each of the first wheel axis and the second wheel axis, and wherein the reference plane extends through the opening in the housing when the band saw is in at least the lowered position.

[0005] In some aspects, the techniques described herein relate to a benchtop band saw including: a base; a clamp assembly coupled to the base and configured to hold a workpiece; and a band saw moveably coupled to the base between a first position and a second position, the band saw including a housing having a first end moveably coupled to the base, an opposite, second end, and an opening in which the workpiece is at least partially receivable during a cutting operation, the opening being located between the first end and the second end, a motor positioned within the housing, a first wheel supported by the housing for rotation about a first wheel axis, the first wheel being coupled to the motor to receive torque therefrom, a second wheel supported by the housing for rotation about a second wheel axis, a band saw blade extending around the first wheel and the second wheel and traveling in a continuous loop around the first wheel and the second wheel, the band saw blade traversing the opening in the housing and configured to cut the workpiece as the band saw moves from the first position to the second position, a handle coupled to the housing proximate the first end, the handle having a mount portion fixed to the housing and a grip portion extending transverse to the mount portion, the grip portion including a rear surface facing the first end of the housing, and a trigger protruding from the rear surface of the grip portion of the handle, the trigger being configured to actuate the motor when depressed.

[0006] In some aspects, the techniques described herein relate to a benchtop band saw including: a base; a clamp assembly coupled to the base and configured to hold a workpiece; and a band saw

moveably coupled to the base, the band saw including a housing having a first end moveably coupled to the base, an opposite, second end, and an opening in which the workpiece is at least partially receivable during a cutting operation, the opening being located between the first end and the second end, a motor positioned within the housing, a first wheel supported by the housing for rotation about a first wheel axis, the first wheel being coupled to the motor to receive torque therefrom, a second wheel supported by the housing for rotation about a second wheel axis, a band saw blade extending around the first wheel and the second wheel and traveling in a continuous loop around the first wheel and the second wheel, the band saw blade traversing the opening in the housing and configured to cut the workpiece, and a blade guard movably coupled to the housing between a first position and a second position, wherein the blade guard is preloaded against the housing to lock the blade guard in a desired position when the blade guard is motionless.

[0007] In some aspects, the techniques described herein relate to a benchtop band saw including: a base; a clamp assembly coupled to the base and configured to hold a workpiece; and a band saw moveably coupled to the base, the band saw including a housing having a first end moveably coupled to the base, an opposite, second end, and an opening in which the workpiece is at least partially receivable during a cutting operation, the opening being located between the first end and the second end, a motor positioned within the housing, a drive wheel assembly rotationally driven by the motor, the drive wheel assembly including a drive wheel and a driven wheel, a band saw blade extending around the drive wheel assembly and traveling in a continuous loop around the drive wheel assembly, the band saw blade configured to pass through the opening of the housing to cut the workpiece, and a blade guide assembly coupled to the housing and configured to support the band saw blade as the band saw blade passes through the opening, the blade guide assembly including a first bearing set disposed on a first end of the opening and a second bearing set disposed on a second end of the opening, wherein the first bearing set includes a first bearing and a second bearing separated from each other by a gap within which the band saw blade is supported, and wherein the first bearing is adjustable relative to the second bearing to adjust a width of the gap.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a front perspective view of a benchtop band saw according to an embodiment of the present invention.

[0009] FIG. 2 is a rear perspective view of the benchtop band saw of FIG. 1.

[0010] FIG. 3 is a front perspective view of the benchtop band saw of FIG. 1, showing a band saw with portions removed.

[0011] FIG. 4 is a back perspective view of the benchtop band saw of FIG. 1, showing a band saw with portions removed.

[0012] FIG. 5 is a front perspective view of a handle of the benchtop band saw of FIG. 1.

[0013] FIG. 6 is a cross-sectional view of the benchtop band saw of FIG. 1, showing a drive wheel assembly.

[0014] FIG. 7 is a back perspective view of the benchtop band saw of FIG. 1, showing a table with portions removed.

[0015] FIG. 8 is a front perspective view of a table of a benchtop band saw according to another embodiment of the invention.

[0016] FIG. 9 is a side view of the benchtop band saw of FIG. 1, showing the benchtop band saw moving from a raised position to a lowered position.

[0017] FIG. 10 is a front perspective view of a band saw according to another embodiment of the invention.

[0018] FIG. **11** is a rear perspective view of the band saw of FIG. **10**.

[0019] FIG. **12** is an enlarged view of the band saw of FIG. **10** illustrating a blade guard disposed in a first position.

[0020] FIG. **13** is a front perspective view of the blade guard of FIG. **12**.

[0021] FIG. **14** is a rear perspective view of the blade guard of FIG. **12**.

[0022] FIG. **15** is a cross-sectional view of the blade guard of FIG. **12** through section line **15-15** in FIG. **10**, the blade guard is coupled to a housing of the band saw of FIG. **10**.

[0023] FIG. **16** is an enlarged view of the band saw of FIG. **10**, illustrating a blade guide assembly.

[0024] FIG. **17** is a perspective view of a first bearing set of the blade guide assembly of FIG. **16**.

[0025] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

[0026] FIGS. **1** and **2** illustrate a benchtop band saw **10** including a band saw **11** pivotably coupled to a table **12**. The band saw **11** is configured to pivot between a raised position and a lowered position for performing a cutting operation (shown in FIG. **9**) on a workpiece.

[0027] FIG. **3** illustrates a band saw **11** including a frame or housing **14** supporting a drive mechanism **16** (FIG. **6**). The drive mechanism **16** includes a motor **18** and a transmission **22**. In some embodiments of the band saw **11**, the motor **18** is configured as a brushless DC (BLDC) electric motor and the band saw **11** includes a removable, rechargeable battery pack **26** (FIG. **3**) for supplying power to the motor **18**. In other embodiments, the rechargeable battery pack **26** may be located elsewhere in the benchtop band saw **11**, such as on the table **12**. With reference to FIG. **4**, the drive mechanism **16** further includes a drive wheel assembly **30**, which is driven by the motor **18** via the transmission **22**. In the illustrated embodiment, the motor **18**, the transmission **22**, and the drive wheel assembly **30** are supported by the housing **14**, but may be elsewhere in other embodiments. The transmission **22** may include different gear train arrangements configured to provide a low-speed, high-torque output from the high-speed, low-torque input of the motor **18**. The drive mechanism **16** is operable to drive a continuous band saw blade **34** to cut a workpiece.

[0028] With reference to FIGS. **3** and **4**, the housing **14** extends between a first end **35** that is pivotably coupled to the table **12** and an opposite, second end **36**. The housing **14** includes a handle **38** coupled to the housing **14** proximate the second end **36**. In some embodiments, the handle **38** is a D-shaped handle with a trigger **40** disposed on an inner surface of the handle **38**. As shown in FIG. **5**, in other embodiments, the handle **38** may be a pommel handle **38** having a mount portion **41** fixed to the housing **14** and a grip portion **42** extending transverse to the mount portion **41**. The grip portion **42** includes a rear surface facing the first end **35** of the housing **14**. The handle **38** is provided with a trigger **44** to selectively supply power to the band saw **11**. The trigger **44** protrudes from the rear surface of the grip portion **42** of the pommel handle **38**. When the trigger **44** is depressed, the motor **18** is activated. In the illustrated embodiment of the band saw **11**, the battery pack **26** is supported by the housing **14** between the first end **35** and the second end **36**. In other embodiments, the battery pack **26** may be supported elsewhere on the housing **14** or offboard the housing **14**, including on the table **12**.

[0029] With reference to FIG. **3**, the housing **14** also includes a deck **46** and a guard **50** coupled to the deck **46**. The deck **46** and the guard **50** form a cavity **54** (FIG. **4**) in which the drive wheel assembly **30** and the band saw blade **34** are at least partially positioned. The housing **14** also defines an opening **58** disposed between the first end **35** of the housing **14** and the second end **36** of the housing **14**. In some embodiments, the opening **58** is positioned at a center of the housing **14**. In other embodiments, the opening **58** is offset from the center of the housing **14**. The deck **46** and the

guard **50** substantially cover the band saw blade **34** when the band saw blade **34** is in a shielded position (i.e., when the band saw blade **34** is not within the opening **58**; FIG. **4**). However, the deck **46** and the guard **50** do not cover the band saw blade **34** when the band saw blade **34** is in an exposed position (i.e., when the band saw blade **34** performs the cutting operation). In the exposed position, the band saw blade **34** is fully exposed and unobstructed by the guard **50**, allowing workpieces to be cut when entering the opening **58**. In other words, the band saw blade **34** traverses the opening **58** such that the workpiece is partially received the opening **58** during the cutting operation.

[0030] With reference to FIG. **4**, the drive wheel assembly **30** includes a drive wheel **70** that is driven by the motor **18** and a driven wheel **74** that is driven by the drive wheel **70** via the band saw blade **34**. Specifically, the band saw blade **34** extends around the drive wheel **70** and the driven wheel **74** such that rotation of the drive wheel **70** rotates the band saw blade **34**, which thereby rotates the driven wheel **74**. Portions of the drive wheel **70** and the driven wheel **74** are covered by the deck **46** and the guard **50**. The opening **58** is positioned between the drive wheel **70** and the driven wheel **74** such that the band saw blade **34** traverses the opening **58**. The drive wheel **70** rotates about a drive wheel axis **78** defined by a drive wheel spindle **82** (FIG. **6**). The driven wheel **74** rotates about a driven wheel axis **79** (FIG. **4**) that is parallel with the drive wheel axis **78**.

[0031] At one end, the drive wheel spindle **82** is coupled for co-rotation with a carrier **62** of the transmission **22** which, in the illustrated embodiment of the band saw **11**, includes a multi-stage planetary gear train. At the other end, the drive wheel spindle **82** is keyed to the drive wheel **70**, such that the drive wheel spindle **82** and the drive wheel **70** co-rotate. In other embodiments, a third wheel may be positioned between the drive wheel spindle **82** and the drive wheel **70** for co-rotation therewith to increase the cutting capacity of the band saw **11**. The drive wheel spindle **82** also extends from the transmission **22** through an aperture **66** in the deck **46**. The aperture **66** is diametrically larger than the drive wheel spindle **82**, such that the aperture **66** does not engage or support the drive wheel spindle **82**. The drive wheel spindle **82** is axially supported by a thrust bearing **84** adjacent the carrier **62**. Specifically, the thrust bearing **84** engages and axially supports a flange **86** extending radially outward from the drive wheel spindle **82**. The drive wheel spindle **82** is radially supported by a roller bearing **83** disposed adjacent the aperture **66**. The flange **86** is disposed between the carrier **62** and the thrust bearing **84**. Therefore, the drive wheel **70** is drivingly connected to the motor **18** via the transmission **22** and the drive wheel spindle **82**. In the illustrated embodiment of the band saw **11**, a rotational axis **80** of the motor **18** is coaxially aligned with the drive wheel axis **78**. In other embodiments, the motor rotational axis **80** and the drive wheel axis **78** may be parallel or at some other oblique angle relative to each other. A tire **88** is coupled to the outer peripheral surface of each of the wheels **70**, **74** (although only one of which is shown). Each tire **88** grips the band saw blade **34** to drive motion thereof.

[0032] With reference to FIGS. **1** and **2**, the band saw **11** is pivotably coupled to the table **12** by a support arm **90**, as explained in greater detail below. The band saw **11** is pivotably coupled to the support arm **90** about a chop axis **94** (FIG. **2**) defined between the raised position and a lowered position for performing the cutting operation on the workpiece, as shown in FIG. **9**. In the raised position, the band saw **11** is angled relative to the table **12**. In the lowered position, the band saw **11** is parallel with the table **12**.

[0033] With reference to FIG. **7**, the table **12** includes a base **98** and a clamp assembly **102** affixed to the base **98**. The base **98** is configured to act as a stand for the band saw **11**. More specifically, the band saw **11** is supported by the base **98** during the cutting operation. The base **98** includes mounting apertures **106** for optionally securing the base **98** a workbench. In some embodiments, the mounting apertures **106** are disposed at corners of the base **98**. In other embodiments, the mounting apertures **106** may be disposed at alternative locations in the base **98**. The table **12** additionally includes a handle **110** for carrying the benchtop bandsaw **10**. In the illustrated embodiment, the handle **110** is disposed at a side of the base **98** opposite the clamp assembly **102**.

In some embodiments, the base **98** may include measurement indicia, such as a miter angle scale **111** and a material cut-off scale **112**, located on a top surface **114** of the base **98** (FIG. 8). In some embodiments, the base **98**, the handle **110**, and a portion of the clamp assembly **102** may be integrally formed as a single piece.

[0034] The top surface **114** of the base **98** defines a base plane. The band saw **11** is angled relative to the base plane. Specifically, drive wheel axis **78** and the driven wheel axis **79** are oriented at an oblique angle relative to the base plane (FIG. 2). In some embodiments, the drive wheel axis **78** and the driven wheel axis **79** may be angled at 45 degrees relative to the base plane. In other embodiments, the drive wheel axis **78** and the driven wheel axis **79** may be angled at between 45 degrees and 90 degrees relative to the base plane. In further embodiments, the drive wheel axis **78** and the driven wheel axis **79** may be angled at between 0 degrees and 45 degrees relative to the base plane.

[0035] With reference to FIG. 7, the clamp assembly **102** holds the workpiece in place relative to the table **12** during a cutting operation. The clamp assembly **102** includes a clamp base **120**, a first, movable jaw **124** slidable relative to the clamp base **120**, and a second, stationary jaw **128**. In some embodiments, the clamp base **120** includes mounting holes through which fasteners are inserted to secure the clamp base **120** to the base **98**. In other embodiments, such as that shown in FIG. 8 the clamp base **120** may be integrally formed with the base **98** (e.g., by a casting process). With reference to FIG. 7, the clamp assembly **102** also includes a rotatable screw **132**. The rotatable screw **132** is received in a threaded aperture in the clamp base **120**. Rotating the screw **132** slides the first jaw **124** along a surface of the clamp base **120** toward or away from the second jaw **128** to adjust a clamping force applied to a workpiece between the jaws **124**, **128**. The second jaw **128** includes a flat surface **136** against which the workpiece is held when the workpiece is being clamped by the clamp assembly **102**. The flat surface **136** defines a vertical plane **138** that is perpendicular to the base plane. As shown in FIG. 9, the vertical plane **138** extends through the opening **58** in the band saw housing **14** in each of the raised position and the lowered position. In operation, the rotatable screw **132** is rotated such that the first jaw **124** moves away from the second jaw **128**. The workpiece is placed between the first jaw **124** and the second jaw **128**. The rotatable screw **132** is rotated such that the first jaw **124** moves toward the second jaw **128**. Once the first jaw **124** is touching the workpiece and the second jaw **128** is touching the workpiece, the workpiece is held by the clamp assembly **102** such that the cutting operation may be performed.

[0036] The support arm **90** includes a band saw mount **140** pivotably coupled to a support arm base **144** via a pivot joint **148**. In other embodiments, the band saw mount **140** may be slidably coupled to the support arm base **144** such that the band saw **11** may translate relative to the base **98**. In other embodiments, the support arm **90** may include additional components between the base **98** and the band saw **11**. The band saw mount **140** is coupled to the first end **35** of the band saw **11** via fasteners. In the illustrated embodiment, the pivot joint **148** defines the chop axis **94** for the band saw **11**. In other embodiments, the chop axis **94** may be defined elsewhere on the band saw **10**. The chop axis **94** extends parallel to the base **98**. In other embodiments, the chop axis **94** may extend perpendicular to the base **98**. In other embodiments, the chop axis **94** may extend at an alternative angle relative to the base **98**. The pivot joint **148** includes detents that engage with pins to hold the band saw **11** in one or more discrete positions between the raised position and the lowered position shown in FIG. 9, and prevent the band saw **11** from pivoting beyond a predetermined range. Specifically, the pivot joint **148** includes a first detent (not shown) that engages with a hard-stop bolt (not shown) when the band saw **11** is at a 0 degree chop angle relative to the base plane. Therefore, the first detent prevents the band saw **11** from coming into contact with the base **98**. The pivot joint **148** also includes a second detent (not shown) that engages with the hard stop bolt when the band saw **11** is at a 45 degree chop angle relative to the base plane. Therefore, the band saw **11** is solely moveable between 0 degrees and 45 degrees relative to the base plane. When the band saw **11** is at the 0 degree chop angle, the band saw **11** is in the lowered position. When the band saw **11**

is at the 45 degree chop angle, the band saw **11** is in the raised position. In other embodiments, the pivot joint **148** may allow an alternative range of motion for the band saw **11**.

[0037] The second detent additionally may engage with a lock-down pin (not shown) when the band saw **11** is at the 0 degree chop angle relative to the base plane. The lock-down pin holds the band saw **11** at the 0 degree chop angle when a user moves the lock-down pin into the second detent. Therefore, the band saw **11** can be locked at the 0 degree chop angle for transporting or carrying the benchtop bandsaw **10**. The pivot joint **148** also includes a third detent (not shown) that may engage with the lock down pin when the band saw **11** is at a 30 degree chop angle relative to the base plane. The lock-down pin holds the band saw **11** at the 30 degree chop angle when the user moves the lock-down pin into the third detent. This allows the user to lock the band saw **11** in place so that the user can move the workpiece. In other embodiments, the pivot joint **148** may include additional and/or alternative detents for locking the band saw **11** in place. Additionally, the detents may be disposed at alternative locations such that the band saw **11** is held at differing angles than the angles disclosed above.

[0038] With reference to FIG. 7, the support arm base **144** is pivotably coupled to the base **98** about a miter axis **152** that extends perpendicular to the base plane. The miter axis **152** extends perpendicular to the chop axis **94**. The support arm **90** is pivotably coupled to the base **98** via an eccentric pivot (not shown). The eccentric pivot allows the support arm **90** to move linearly along the base plane and to rotate about the miter axis **152**. Specifically, the eccentric pivot allows linear motion of the support arm **90** in a direction perpendicular with the vertical plane **138**. In other words, linear motion of the support arm **90** moves the band saw **11** toward the clamp assembly **102**. In other embodiments, the eccentric pivot may be replaced with a pivot pin that solely allows rotational motion. In such embodiments, the support arm **90** can solely pivot about the miter axis **152**. The miter axis **152** is located within a reference plane oriented parallel to each of the drive wheel axis **78** and the driven wheel axis **79**. Additionally, the reference plane extends through the opening **58** in the housing **14** of the band saw **11**. In other words, the miter axis **152** is proximate a center of the base **98** and a center of the band saw **11**. The support arm **90**, and therefore the band saw **11**, are pivotable about the miter axis **152** between a first miter angle and a second miter angle. In the first miter angle, the band saw **11** is at a first angle relative to the vertical plane **138**. In the second miter angle, the band saw **11** is at a second angle relative to the vertical plane **138**, the first angle is different than the second angle. The vertical plane **138** extends through the opening **58** of the housing **14** in each of the first miter angle and the second miter angle. The vertical plane **138** and the flat surface **136** of the second jaw **128** are generally aligned with a rear surface of the opening **58** for miter angles between the first miter angle and the second miter angle. Therefore, an entire length of the opening **58** is available during the cutting operation, increasing a cutting capacity of the band saw **11**.

[0039] With reference to FIGS. 1-2 and 9, in operation, the user pivots the band saw **11** about the chop axis **94** to the raised position. The user places the workpiece between the first jaw **124** and the second jaw **128** of the clamp assembly **102**, and moves the first jaw **124** toward the second jaw **128** to clamp the workpiece. The user pivots the band saw **11** about the miter axis **152** to a desired angle. The desired angle corresponds to a cut angle for the workpiece. Once the band saw **11** is at the desired angle, the user depresses the trigger **44** to activate the motor **18** and drive the saw blade **34** through the opening **58**. The user moves the band saw **11** from the raised position to the lowered position, performing the cutting operation. Once at the lowered position, further movement of the band saw **11** is inhibited. The user moves the band saw **11** from the lowered position to the raised position and releases the trigger **44**.

[0040] Since the miter axis **152** is proximate the center of the band saw **11**, the miter axis **152** provides a more uniform arc-path. The arc-path limits the cutting capacity of the band saw **11**. Due to the location of the miter axis **152**, the band saw **11** has an improved cutting capacity. For example, when the band saw **11** is pivoted about the miter axis **152** to a 45 degree miter angle (i.e.,

when the saw blade **34** defines a 45 degree angle with the vertical plane **138**), the band saw **11** may cut a pipe having a diameter between 45 mm and 50 mm. When the band saw **11** is pivoted about the miter axis **152** to a 0 degree miter angle (i.e., when the saw blade **34** defines a 90 degree angle with the vertical plane **138**), the band saw **11** may cut a pipe having a diameter between 120 mm and 125 mm. In other embodiments, the cutting capacity of the band saw **11** may differ.

[0041] FIGS. **10-14** illustrate another band saw **211** configured to be pivotably coupled to the table **12** of the benchtop band saw **10** of FIGS. **1** and **2**. The band saw **211** is similar to the band saw **11** of FIGS. **1-4**; therefore, like structure will be identified by like reference numbers plus “**200**.”

[0042] The band saw **211** includes a blade guard **356** movably coupled to a housing **214** of the band saw **211**. More specifically, the blade guard **356** is movably coupled to a deck **246** of the housing **214** proximate a second end **236** of the housing **214**, where a drive wheel **270** of the band saw **211** is located. The blade guard **356** has a body **360** with a slot **364** defined within a top portion of the body **360**, a tab **368** extending from the top portion of the body **360**, and a base **372** integrally formed with a bottom portion of the body **360** (FIGS. **13** and **14**). In the illustrated embodiment, the base **372** of the blade guard **356** extends perpendicular from the body **360** of the blade guard **356**. In other embodiments, the base **372** of the blade guard **356** may extend obliquely from the body **360** of the blade guard **356**. As such, the blade guard **356** is configured to cover or shield a user from a portion of a continuous band saw blade **234** not being used to cut a workpiece during operation of the bandsaw **211**.

[0043] With reference to FIGS. **10**, **12**, and **15**, the blade guard **356** is fastened to the housing **214** of the band saw **211** by two fasteners **376a**, **376b** that extend through the slot **364** of the blade guard **356**. In particular, the fasteners **376a**, **376b** extend through the slot **364** and are threaded to the deck **246**. As such, the blade guard **356** is indirectly coupled to the housing **214** of the band saw **211** by the fasteners **376a**, **376b** and able to move with respect to the housing **214**.

[0044] The blade guard **356** is slidable between a first position (i.e., a retracted position) and a second position (i.e., an extended position). In the first position, as shown in FIG. **12**, the blade guard **356** is adjusted furthest rearward such that the blade guard **356** does not block an opening **258** disposed between a first end **235** of the housing **214** and a second end **236** of the housing **214**. In the second position, the blade guard **356** is adjusted forward such that the blade guard **356** blocks at least a portion of the opening **258**. The blade guard **356** is slidable between the first position and the second position by the user to block varying amounts of the opening **258**. More specifically, the user may grasp the tab **368** to move the blade guard **356** to a desired position in which the blade guard **356** at least partially blocks the opening **258** depending on a size of the workpiece being cut.

[0045] The band saw **211** further includes a biasing member **380** disposed between the blade guard **356** and the fasteners **376a**, **376b** (FIG. **15**). More specifically, the biasing member **380** is disposed between a head **384** of each fastener **376a**, **376b** and a top surface **388** of the blade guard **356**. In some embodiments, the biasing member **380** is a wave spring. In other embodiments, the biasing member **380** is a rubber bushing. The biasing member **380** preloads the blade guard **356** against the deck **246** of the band saw **211** to create friction between the blade guard **356** and the deck **246**, and thereby automatically lock the blade guard **356** in a desired position. To slide the blade guard **356** with respect to the housing **214** of the band saw **211**, the user may grasp the tab **368** to push or pull the blade guard **356** toward the opening **258** or away from the opening **258** with a sufficiently high force that is able to overcome the friction created between the blade guard **356** and the housing **214** by the biasing member **380**.

[0046] In operation, the user pivots the band saw **211** to the raised position. The user then places a workpiece between the first jaw **124** and the second jaw **128** of the clamp assembly **102** and moves the first jaw **124** toward the second jaw **128** to clamp the workpiece. The user grasps the tab **368** of the blade guard **356** to move the blade guard **356** to a desired position between the first position and the second position. Once the user releases the blade guard **356** and the blade guard **356** is

motionless, the friction created between the blade guard **356** and the housing **214** of the band saw **211** by the biasing member **380**, maintains the blade guard **356** in the desired position without requiring any further action from the user to lock the blade guard **356** in place. Further, the user pivots the band saw **211** to a desired angle corresponding to a cut angle for the workpiece. Once the band saw **211** is at the desired angle, the user engages an actuator to activate the motor **218** and drive the saw blade **234** through the opening **258**. The user moves the band saw **211** from the raised position to the lowered position, performing the cutting operation. Once at the lowered position, further movement of the band saw **211** is inhibited. The user moves the band saw **211** from the lowered position to the raised position and disengages the actuator. In some embodiments, the band saw **211** can be used without the table **12**.

[0047] With reference to FIG. **11**, the band saw **211** includes a blade guide assembly **392** having a first bearing set **396a** and a second bearing set **396b** disposed on each side of the opening **258**. The first and second bearing sets **396a**, **396b** engage opposing sides of the saw blade **234** and thereby support the saw blade **234** as the blade **234** enters and exits the opening **258** during operation.

[0048] With reference to FIGS. **16** and **17**, the first bearing set **396a** includes a first mounting block **400a** coupled to the housing **214** of the band saw **211** and a first pair of bearings **404a1**, **404a2** rotatably supported by the first mounting block **400a**. The second bearing set **396b** includes a second mounting block **400b** coupled to the housing **214** of the band saw **211** and a second pair of bearings **404b1**, **404b2** rotatably supported by the second mounting block **400b**. The first mounting block **400a** is disposed proximate a first end of the opening **258** and the second mounting block **400b** is disposed proximate a second end of the opening **258**. A gap **408a**, **408b**, through which the saw blade **234** extends, is respectively defined between the bearings **404a1**, **404a2** of the first bearing set **396a** and the bearings **404b1**, **404b2** of the second bearing set **396b**.

[0049] With reference to FIG. **17**, a first bearing **404a1** of the first bearing set **396a** is adjustable. Specifically, a user may adjust the position of the first bearing **404a1** relative to the first mounting block **400a** for varying the size of the gap **408a** to accommodate saw blades of different thicknesses. The first bearing **404a1** includes an eccentric sleeve **412a** that is rotatable by a user to move the first bearing **404a1** toward or away from the second bearing **404a2** of the first bearing set **396a**. Also, the first bearing **404a1** has a fastener **416a** for axially retaining the first bearing **404a1** to the first mounting block **400a**. More specifically, when the eccentric sleeve **412a** is rotated in a first direction, the first bearing **404a1** is moved away from the second bearing **404a2** to increase the size of the gap **408a**. When the eccentric sleeve **412a** is rotated in a second direction opposite the first direction, the first bearing **404a1** is moved toward the second bearing **404a2** to reduce the size of the gap **408a**. As such, adjusting the position of the first bearing **404a1** relative to the second bearing **404a2** will reduce the occurrence of blade wobbling by the saw blade **234**.

Although the second bearing set **396b** is not shown in FIG. **17**, in some embodiments, the second bearing set **396b** can be similar to the first bearing set **396a** (that is, with one of the bearings **404b1**, **404b2** being adjustable relative to the other to adjust the width of the gap **408b**). The first and second bearing sets **396a**, **396b** can also be incorporated in the band saw **11** of FIGS. **1-9**. In some embodiments, the band saw **11** can be used without the table **12**.

[0050] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

[0051] Various features of the present invention are set forth in the following claims.

Claims

1. A benchtop band saw comprising: a base; a clamp assembly coupled to the base and configured to hold a workpiece; a support arm pivotably coupled to the base about a miter axis that extends perpendicular to the base; and a band saw coupled to the support arm and moveable relative to the

support arm and the base between a raised position and a lowered position for performing a cutting operation on the workpiece, the band saw including a housing defining an opening in which the workpiece is at least partially receivable during the cutting operation, a motor positioned within the housing, a first wheel supported by the housing for rotation about a first wheel axis, the first wheel being coupled to the motor to receive torque therefrom, a second wheel supported by the housing for rotation about a second wheel axis, and a band saw blade extending around the first wheel and the second wheel and traveling in a continuous loop around the first wheel and the second wheel, the band saw blade traversing the opening in the housing, wherein the miter axis is located within a reference plane oriented parallel to each of the first wheel axis and the second wheel axis, and wherein the reference plane extends through the opening in the housing when the band saw is in at least the lowered position.

2. The benchtop band saw of claim 1, wherein the band saw is pivotable about the miter axis between a first miter angle and a different, second miter angle, wherein the clamp assembly includes a stationary jaw defining a vertical plane, and wherein the vertical plane extends through the opening in the housing in at least one of the first miter angle and the second miter angle when the band saw is in each of the raised position and the lowered position.

3. The benchtop band saw of claim 2, wherein the vertical plane extends through the opening in the housing in each of the first miter angle and the second miter angle when the band saw is in each of the raised position and the lowered position.

4. The benchtop band saw of claim 1, wherein the band saw is pivotably coupled to the support arm about a chop axis oriented perpendicular to the miter axis.

5. The benchtop band saw of claim 4, wherein the support arm has a support arm base and a band saw mount pivotably coupled to the support arm base by a pivot joint that defines the chop axis, the support arm base is coupled to the base.

6. The benchtop band saw of claim 1, wherein the miter axis is proximate a center of the band saw.

7. The benchtop band saw of claim 1, wherein the base includes a miter angle scale and a material cut-off scale located on a top surface of the base.

8. A benchtop band saw comprising: a base; a clamp assembly coupled to the base and configured to hold a workpiece; and a band saw moveably coupled to the base between a first position and a second position, the band saw including a housing having a first end moveably coupled to the base, an opposite, second end, and an opening in which the workpiece is at least partially receivable during a cutting operation, the opening being located between the first end and the second end, a motor positioned within the housing, a first wheel supported by the housing for rotation about a first wheel axis, the first wheel being coupled to the motor to receive torque therefrom, a second wheel supported by the housing for rotation about a second wheel axis, a band saw blade extending around the first wheel and the second wheel and traveling in a continuous loop around the first wheel and the second wheel, the band saw blade traversing the opening in the housing and configured to cut the workpiece as the band saw moves from the first position to the second position, a handle coupled to the housing proximate the first end, the handle having a mount portion fixed to the housing and a grip portion extending transverse to the mount portion, the grip portion including a rear surface facing the first end of the housing, and a trigger protruding from the rear surface of the grip portion of the handle, the trigger being configured to actuate the motor when depressed.

9. The benchtop band saw of claim 8, wherein the handle is a pommel handle.

10. A benchtop band saw comprising: a base; a clamp assembly coupled to the base and configured to hold a workpiece; and a band saw moveably coupled to the base, the band saw including a housing having a first end moveably coupled to the base, an opposite, second end, and an opening in which the workpiece is at least partially receivable during a cutting operation, the opening being located between the first end and the second end, a motor positioned within the housing, a first wheel supported by the housing for rotation about a first wheel axis, the first wheel being coupled

to the motor to receive torque therefrom, a second wheel supported by the housing for rotation about a second wheel axis, a band saw blade extending around the first wheel and the second wheel and traveling in a continuous loop around the first wheel and the second wheel, the band saw blade traversing the opening in the housing and configured to cut the workpiece, and a blade guard movably coupled to the housing between a first position and a second position, wherein the blade guard is preloaded against the housing to lock the blade guard in a desired position when the blade guard is motionless.

11. The benchtop band saw of claim 10, wherein the band saw further includes a biasing member configured to preload the blade guard against the housing to lock the blade guard in the desired position.

12. The benchtop band saw of claim 11, wherein the band saw further includes a fastener configured to fasten the blade guard to the housing of the band saw, and wherein the biasing member is disposed between the fastener and the blade guard.

13. The benchtop band saw of claim 12, wherein the biasing member is a wave spring.

14. The benchtop band saw of claim 12, wherein the biasing member is a rubber bushing.

15. The benchtop band saw of claim 10, wherein in the first position, the blade guard is positioned with the opening being unblocked, and wherein in the second position, the blade guard is positioned with at least a portion of the opening being blocked by the blade guard.

16. A benchtop band saw comprising: a base; a clamp assembly coupled to the base and configured to hold a workpiece; and a band saw moveably coupled to the base, the band saw including a housing having a first end moveably coupled to the base, an opposite, second end, and an opening in which the workpiece is at least partially receivable during a cutting operation, the opening being located between the first end and the second end, a motor positioned within the housing, a drive wheel assembly rotationally driven by the motor, the drive wheel assembly including a drive wheel and a driven wheel, a band saw blade extending around the drive wheel assembly and traveling in a continuous loop around the drive wheel assembly, the band saw blade configured to pass through the opening of the housing to cut the workpiece, and a blade guide assembly coupled to the housing and configured to support the band saw blade as the band saw blade passes through the opening, the blade guide assembly including a first bearing set disposed on a first end of the opening and a second bearing set disposed on a second end of the opening, wherein the first bearing set includes a first bearing and a second bearing separated from each other by a gap within which the band saw blade is supported, and wherein the first bearing is adjustable relative to the second bearing to adjust a width of the gap.

17. The benchtop band saw of claim 16, wherein the first bearing includes an eccentric sleeve configured to be rotated for moving the first bearing relative to the second bearing.

18. The benchtop band saw of claim 17, wherein the eccentric sleeve is rotatable between a first direction, in which the first bearing is moved toward the second bearing, and a second direction, in which the first bearing is moved away from the second bearing.

19. The benchtop band saw of claim 16, wherein the gap is a first gap, wherein the second bearing set includes a first bearing and a second bearing separated from each other by a second gap within which the band saw blade is supported, and wherein the first bearing of the second bearing set is adjustable relative to the second bearing of the second bearing set to adjust a width of the second gap.

20. The benchtop band saw of claim 19, wherein each of the first and second bearing sets further includes a mounting block coupled to the housing of the band saw and configured to rotatably support the first bearing and the second bearing of the respective bearing sets.
